MUTUAL FUND ANALYTICS

Mutual Fund Analytics

A comprehensive analytics project leveraging Python, SQL, and Power BI to transform raw mutual fund data into actionable investment intelligence for fund managers and analysts.

Problem Statement

Investors lack accessible tools to evaluate fund performance, compare schemes, and understand behavioural trends. This project bridges that gap with data-driven insights.

Project Scope

From data ingestion and engineering to advanced risk analytics and interactive dashboards — covering the full analytical lifecycle for mutual fund evaluation.

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Analyse NAV Trends

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Evaluate Fund Performance

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Study Investor Behaviour

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Build Interactive Dashboard

Technologies Used

A modern, open-source stack spanning data manipulation, visualisation, database engineering, and business intelligence — selected for scalability and analytical depth.

Category	Tools	Role
Data Processing	Python, Pandas, NumPy	Wrangling & computation
Visualisation	Matplotlib, Seaborn, Plotly	Static & interactive charts
Database	SQLite	Storage & ORM layer
BI & Reporting	Power BI	Interactive dashboards
Environment & VCS	Vs code, Git & GitHub	Development & versioning

Data Collection

Nine structured datasets form the foundation of this project, covering fund metadata, NAV history, investor transactions, benchmarks, and industry-level aggregates.

fund_master

Fund details — scheme codes, categories, and AMC metadata.

nav_history

Daily Net Asset Value records for trend analysis.

investor_transactions

SIP and lumpsum investment flows by investor.

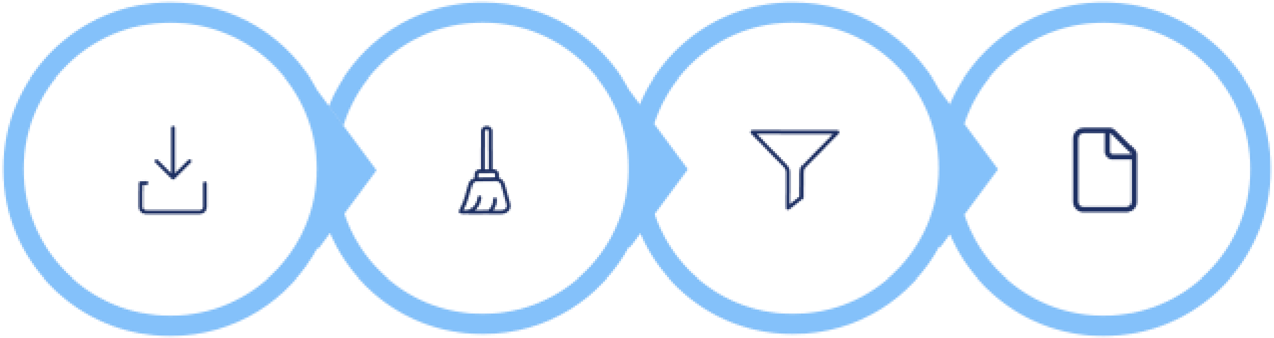
scheme_performance

Returns and risk metrics per scheme.

Dataset	Description
benchmark_indices	Nifty benchmarks for comparison
portfolio_holdings	Sector allocation per fund
monthly_sip_inflows	SIP statistics over time
category_inflows	Category-level inflow data
industry_folio_count	Industry-wide folio counts
aum_by_fund_house	AUM per asset management company

Data Engineering

A robust ETL pipeline ensures data integrity from raw ingestion through to the analytics-ready SQLite warehouse. Each stage is designed for reproducibility and auditability.



Ingest CSVs

Clean Data

Remove Duplicates

Load SQLite

The pipeline processes nine source datasets through validation, transformation, and loading — producing a normalised relational schema accessible via SQLAlchemy ORM.

Exploratory Data Analysis

EDA uncovers structural patterns across NAV trajectories, asset flows, investor demographics, and portfolio composition — guiding downstream modelling decisions.

NAV & AUM Trends

Time-series analysis of fund valuations and assets under management.

SIP Inflow Trends

Monthly SIP contribution patterns and seasonality.

Correlation Matrix

Heatmap of returns correlations across fund categories.

Geographic Analysis

Investor distribution and folio concentration by region.

Sector Allocation

Portfolio holdings breakdown by industry sector.

Investor Demographics

Behavioural segmentation of investor cohorts.

Performance Analytics

Quantitative metrics evaluate risk-adjusted returns, benchmark alignment, and downside protection — culminating in a composite Fund Scorecard for scheme comparison.

Return Metrics

- Daily Returns
- CAGR (Compound Annual Growth Rate)
- Alpha — excess return over benchmark

Risk Metrics

- Sharpe Ratio — risk-adjusted return
- Sortino Ratio — downside-adjusted return
- Beta — market sensitivity
- Maximum Drawdown
- Tracking Error

The Fund Scorecard aggregates all metrics into a single comparative ranking, enabling quick identification of top-performing schemes.

Interactive Dashboard

A four-page Power BI dashboard delivers self-service analytics with drill-through navigation, dynamic tooltips, and KPI cards for at-a-glance monitoring.

Page 1 — Overview

KPI cards, AUM trends, and category-level inflow summaries.

Page 2 — Fund Performance

Scorecards, risk metrics, and benchmark comparisons.

Page 3 — Investor Behaviour

SIP trends, cohort analysis, and demographic breakdowns.

Advanced Analytics

Beyond standard metrics, the project applies institutional-grade risk models and behavioural analytics to uncover deeper investment insights.

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Historical VaR & CVaR

Value-at-Risk and Conditional VaR quantify tail-risk exposure at 95% and 99% confidence intervals.

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Rolling Sharpe Ratio

Time-varying risk-adjusted performance reveals stability across market regimes.

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Cohort & SIP Continuity

Investor retention by entry cohort and SIP persistence rates over time.

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Recommender & HHI

Fund recommendation engine and Herfindahl-Hirschman Index for portfolio concentration analysis.

Key Findings

The analysis surfaced five headline insights that shape the project's conclusions and inform investor decision-making.

1. **SBI Mutual Fund**

   Recorded the largest AUM among all fund houses analysed.

2. **Large-Cap Stability**

   Large-cap funds generated the most stable returns across categories.

3. **SIP Record High**

   SIP inflows reached all-time highs in December 2025.

4. **2022 Cohort Leadership**

   Investors who began in 2022 invested the highest cumulative amounts.

5. **Diversification Edge**

   Diversified funds showed lower concentration than thematic funds.

Conclusion & Future Enhancements

All four objectives were achieved: NAV trends analysed, performance evaluated, investor behaviour studied, and an interactive dashboard delivered — producing a comprehensive analytics platform.

Project Outcomes

- Robust ETL pipeline with SQLite backend
- Risk-adjusted fund scorecards
- Four-page interactive Power BI dashboard
- Advanced risk and behavioural models

Future Enhancements

- Live NAV API integration
- Streamlit web application
- Monte Carlo simulation
- Efficient Frontier optimisation
- Automated email reporting